2025

COMPUTER SCIENCE AND ENGINEERING

Paper : CSE-1002

(Internet of Things)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1, 2** and ***any four*** questions from the rest of the questions.

1. Answer ***any five*** questions : 2×5

(a) "Cyclic schedulers are more proficient than the table-driven scheduler". — Critically comment with justification.

(b) Assume $p_i = D_i$ for each i. What is the schedulability condition of the Earliest Deadline First algorithm for this system? Is this a necessary and sufficient condition?

(c) Explain how non-real-time tasks might disrupt the operation of real-time tasks running on the same system.

(d) What are the essential components of the Internet of Things?

(e) What is MQTT?

(f) Compare and contrast 'sensors and transducers'.

(g) What is the difference between the Internet of Things (IoT) and Machine-to-Machine (M2M)?

(h) What techniques are used to reduce power consumption in IoT devices?

2. Answer ***any five*** questions : 4×5

(a) Critically comment on the desirable characteristics of a real-time system.

(b) What do you mean by the scheduling point of a task scheduling algorithm? How are scheduling points determined in clock-driven, event-driven and hybrid schedulers?

(c) What do you mean by fault tolerance? Compare and contrast hardware and software fault tolerance.

(d) Discuss and differentiate Arduino Uno and Raspberry Pi boards.

(e) Briefly explain the alternative technologies available for temperature sensors.

(f) Critically comment on any communication technology for IoT application development.

(g) Briefly discuss different types of actuators.

(h) Comment on the architectural layers of an IoT system.

3. The periodic tasks (3, 1), (4, 2) and (6, 1) are scheduled according to the rate monotonic algorithm.
   (a) Draw the time-demand functions of the tasks.
   (b) Are the tasks schedulable? Why or why not?
   (c) Can this graph be used to determine whether the tasks are schedulable according to an arbitrary priority-driven algorithm? Explain your answer. 3+3+4

4. (a) For the following set of jobs with associated parameters, find the job schedule using the priority ceiling protocol :

Job	Release Time	Execution Time	Priority	Critical Section
J1	7	3	1	[Shaded;1]
J2	5	3	2	[Black;1]
J3	4	2	3	
J4	2	6	4	[Shaded; 4 [Black; 1.5]]
J5	0	6	5	[Black; 4]

   (b) Explain how PCP can avoid deadlock, unbounded priority inversions and chain blocking. 7+3

5. (a) What are the main components contributing to IoT device power consumption?
   (b) How does duty cycling improve battery life in IoT systems?
   (c) Write down the steps to calculate the power consumption of an IoT application with an example. 3+3+4

6. (a) Why is a general-purpose OS not suitable for the microcontroller board of an IoT application?
   (b) What are the requirements for an IoT OS for microcontrollers?
   (c) Compare and contrast different OSs for IoT applications. 2+3+5

7. (a) What is an information exchange pattern and why is it needed?
   (b) Briefly explain the different categories of Information Exchange Patterns for IoT applications. 3+7

8. (a) What is a gateway in an IoT system and what are its functional components?
   (b) Briefly explain functional components of the generic M2M architecture. 3+7